The 3-qubit, 2-cavity Hamiltonian

Matched $\chi_{qA} = \chi_{qB}$ JP gate and unitless Lindblad rates

`qumode_sep` technical note

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Abstract

Laboratory Hamiltonian of two storage cavities A, B and three transmons d, q, e; conversion of device frequencies into the unitless rates used by a Lindblad solver; and the joint-parity (JP) gate as a finite-time evolution under a matched shared dispersive coupling.

1 Hardware

Five modes in a chain d–A–q–B–e (Fig. 1). Cavities A, B are the qumodes. Transmons d, e are local ECD ancillas. Transmon q is the shared coupler. This is the two-cavity extension of the one-qumode device in Dutta *et al.* [1] and the Alice/Bob layout of Gao *et al.* [3].

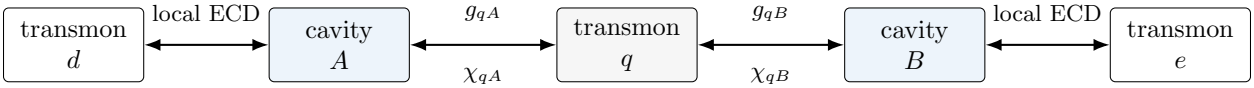


Figure 1: Three transmons and two cavities. For a single-wait JP we set $\chi_{qA} = \chi_{qB}$.

2 Laboratory Hamiltonian

Let $\hat{a}$ ($\hat{b}$) annihilate a photon in A (B), and let $\hat{c}_d, \hat{c}_q, \hat{c}_e$ lower the transmons. Rotating-wave form:

$$\begin{aligned}\frac{\hat{H}}{\hbar} = {} & \omega_A \hat{a}^\dagger \hat{a} + \omega_B \hat{b}^\dagger \hat{b} + \sum_{j\in\{d,q,e\}} \Big[\omega_j \hat{c}_j^\dagger \hat{c}_j + \tfrac{\alpha_j}{2} \hat{c}_j^\dagger \hat{c}_j (\hat{c}_j^\dagger \hat{c}_j - 1)\Big] \\ & + g_{dA}(\hat{a}^\dagger \hat{c}_d + \hat{a}\hat{c}_d^\dagger) + g_{eB}(\hat{b}^\dagger \hat{c}_e + \hat{b}\hat{c}_e^\dagger) \\ & + g_{qA}(\hat{a}^\dagger \hat{c}_q + \hat{a}\hat{c}_q^\dagger) + g_{qB}(\hat{b}^\dagger \hat{c}_q + \hat{b}\hat{c}_q^\dagger) + \hat{H}_{\text{drive}}(t)/\hbar. \end{aligned} \tag{1}$$

Direct d–B, e–A, and d–e couplings are dropped. Cavities are detuned by ~ 1 GHz so they do not hybridize [3].

Far off resonance, $\Delta_{j\mu} = \omega_j - \omega_\mu$, each transmon–cavity pair has a dispersive shift $\chi_{j\mu} = -2g_{j\mu}^2 \alpha_j / [\Delta_{j\mu}(\Delta_{j\mu} + \alpha_j)]$, positive for $\alpha_j < 0$ and $\Delta_{j\mu} > 0$ (transmon above cavity), matching the $-\chi_{j\mu}\hat{n}_\mu |e\rangle\langle e|_j$ sign convention below.

In the lab, χ_{qA} is a measured frequency difference, not a derived symbol. Probe cavity A with a weak tone: the resonance peak sits at one frequency when q is in $|g\rangle$ and at another when q is in $|e\rangle$. That peak-to-peak gap is $\chi_{qA}/2\pi$ (typically quoted in MHz). Equivalently, park a photon in

A and spectroscopy q: the transmon line splits into a ladder $\omega_q,\ \omega_q - \chi_{qA},\ \omega_q - 2\chi_{qA}, \ldots$ indexed by photon number n_A (number splitting). Either way, what moved is a resonance you read on a network analyzer or a qubit spectroscopy scan. The same measurement on q–B, d–A, and e–B gives χ_{qB}, χ_{dA}, and χ_{eB}.

$$\begin{aligned}\frac{\hat{H}_{\text{disp}}}{\hbar} = {} & \widetilde{\omega}_A \hat{n}_A + \widetilde{\omega}_B \hat{n}_B + \sum_{j\in\{d,q,e\}} \widetilde{\omega}_j |e\rangle\langle e|_j \\ & - \chi_{dA}\, \hat{n}_A |e\rangle\langle e|_d - \chi_{eB}\, \hat{n}_B |e\rangle\langle e|_e - \chi_{qA}\, \hat{n}_A |e\rangle\langle e|_q - \chi_{qB}\, \hat{n}_B |e\rangle\langle e|_q \\ & - \tfrac{K_A}{2} \hat{n}_A(\hat{n}_A - 1) - \tfrac{K_B}{2} \hat{n}_B(\hat{n}_B - 1) - K_{AB} \hat{n}_A \hat{n}_B, \end{aligned} \tag{2}$$

with $\hat{n}_A = \hat{a}^\dagger \hat{a}$, $\hat{n}_B = \hat{b}^\dagger \hat{b}$. The $\widetilde{\omega}$ are Lamb-shifted bare frequencies. They imprint ordinary dynamical phases $e^{-i\widetilde{\omega}_A n \tau}$ on Fock states; Kerr imprints extra n-dependent phases. Both are kept in the Lindblad generator below.

Matched shared coupling. JP is a wait under $-\chi_{qA}\hat{n}_A|e\rangle\langle e|_q - \chi_{qB}\hat{n}_B|e\rangle\langle e|_q$. One wait of duration τ maps joint parity onto q iff $\chi_{qA}\tau$ and $\chi_{qB}\tau$ are both odd multiples of π. The easy choice is to set them equal,

$$\chi_{qA} = \chi_{qB}, \qquad \tau = \pi/\chi_{qA}. \tag{3}$$

No extra symbol is needed: after this identification the shared term is $-\chi_{qA}(\hat{n}_A + \hat{n}_B)|e\rangle\langle e|_q$. Because $\omega_A \neq \omega_B$, equal $\chi_{qA} = \chi_{qB}$ generally means *unequal* g_{qA}, g_{qB}. Unmatched shared shifts (Wang: 0.71 vs 1.41 MHz [4]) need two different waits; we do not use that here.

The local shifts χ_{dA} and χ_{eB} are *not* used by JP. They set the ECD rate on (d, A) and (e, B) and may be chosen independently of χ_{qA} and of each other (Eickbusch ECD: $\chi_{dA}/2\pi = 33$ kHz while $\chi_{qA}/2\pi \sim 1$ MHz). They still act during the JP wait as spectators $-\chi_{dA}\hat{n}_A|e\rangle\langle e|_d - \chi_{eB}\hat{n}_B|e\rangle\langle e|_e$. If d or e is in superposition those terms add extra phases; echo $X_d X_e$ at the midpoint, or leave them in the Lindblad Hamiltonian and accept the error.

3 From SI frequencies to unitless numbers

A Lindblad solver takes $\hbar = 1$ and a Hamiltonian whose eigenvalues are inverse times. Pick one laboratory rate as the unit. Here that unit is the matched shared shift χ_{qA} (with $\chi_{qA} = \chi_{qB}$, so $\chi_{qA} = 1$ in code).

Recipe. For every laboratory angular frequency Ω (already in rad/s, i.e. 2π times the quoted Hz), store

$$\widetilde{\Omega} = \Omega/\chi_{qA}. \tag{4}$$

Laboratory time t becomes $\tilde{t} = \chi_{qA} t$. Jump rates follow the same rule: $\tilde{\kappa} = \kappa/\chi_{qA}$. Dimensionless noise over a gate of duration τ is the product $\kappa\tau = \tilde{\kappa}\,\tilde{\tau}$, which is the parameter used by Dutta *et al.* [1].

From Eickbusch [2] and Wang [4]:

quantity	laboratory	unit $\chi_{qA}/2\pi = 1$ MHz	notes
$\chi_{qA} = \chi_{qB}$	1 MHz (choose matched)	1	time unit $1/\chi_{qA} \approx 0.16\ \mu$s
$\chi_{dA} = \chi_{eB}$	33 kHz [2]	0.033	local ECD, kept weak
$\kappa_A = \kappa_B = 1/T_{1,\text{cav}}$	$T_{1,\text{cav}} = 436\ \mu$s	$\tilde{\kappa} = 3.65 \times 10^{-4}$	$\kappa/2\pi \simeq 365$ Hz [1]
$\gamma_q = 1/T_{1,q}$	$T_{1,q} \approx 50\ \mu$s	$\tilde{\gamma}_q = 3.18 \times 10^{-3}$	transmon relaxation
$\gamma_{\phi,q}$ from $T_2 \approx 30\ \mu$s	$T_2^{-1} - T_1^{-1}/2$	$\tilde{\gamma}_{\phi,q} \approx 3.7 \times 10^{-3}$	extra dephasing of q

Conversion of the cavity row, written out:

$$\chi_{qA} = 2\pi \times 10^6 \text{ s}^{-1}, \quad \kappa = \frac{1}{436\ \mu\text{s}} \simeq 2.29 \times 10^3 \text{ s}^{-1}, \quad \tilde{\kappa} = \frac{\kappa}{\chi_{qA}} = 3.65 \times 10^{-4}. \tag{5}$$

A wait $\tau = \pi/\chi_{qA}$ then has $\kappa\tau = \pi\tilde{\kappa} \simeq 1.15 \times 10^{-3}$, the same 10^{-3} scale as the per-ECD-block $\kappa\tau \sim 1.5 \times 10^{-3}$ in Ref. [1].

4 JP gate

The joint-parity operator on the two cavities is

$$\hat{\Pi}_{AB} = (-1)^{\hat{n}_A + \hat{n}_B} = \hat{\Pi}_A \hat{\Pi}_B, \qquad \hat{\Pi}_A = (-1)^{\hat{n}_A}, \quad \hat{\Pi}_B = (-1)^{\hat{n}_B}. \tag{6}$$

It is diagonal in the Fock basis. On a product number state $|n, m\rangle \equiv |n\rangle_A \otimes |m\rangle_B$,

$$\hat{\Pi}_{AB}|n, m\rangle = (-1)^{n+m}|n, m\rangle = \begin{cases} +|n, m\rangle & n + m \text{ even}, \\ -|n, m\rangle & n + m \text{ odd}. \end{cases} \tag{7}$$

Equivalently, the four parity sectors are

parity of n	parity of m	$n + m$	$\hat{\Pi}_{AB}$ eigenvalue
even	even	even	$+1$
even	odd	odd	-1
odd	even	odd	-1
odd	odd	even	$+1$

The odd-joint-parity projector is the $+1$ eigenspace of $(\hat{\mathbb{I}} - \hat{\Pi}_{AB})/2$:

$$\hat{P}_{\text{odd}} = \frac{\hat{\mathbb{I}} - \hat{\Pi}_{AB}}{2}, \qquad \hat{P}_{\text{odd}}|n, m\rangle = \begin{cases} 0 & n + m \text{ even}, \\ |n, m\rangle & n + m \text{ odd}. \end{cases} \tag{8}$$

The JP gate is a $\pi/2$ phase on that subspace,

$$\boxed{\hat{U}_{\text{JP}} = \exp\big[-i(\pi/2)\hat{P}_{\text{odd}}\big] = e^{-i\pi/4} \exp\big[+i(\pi/4)\hat{\Pi}_{AB}\big],} \tag{9}$$

so

$$\hat{U}_{\text{JP}}|n, m\rangle = (-i)^{(n+m) \bmod 2}|n, m\rangle = \begin{cases} +|n, m\rangle & n + m \text{ even}, \\ -i|n, m\rangle & n + m \text{ odd}. \end{cases} \tag{10}$$

Time evolution that realizes it. Initialize q in $|g\rangle$. With $\chi_{qA} = \chi_{qB}$ the shared term in Eq. (2) is $\hat{H}_{\text{JP}}/\hbar = -\chi_{qA}(\hat{n}_A + \hat{n}_B)|e\rangle\langle e|_q$. In unitless time ($\chi_{qA} = 1$), $\tilde{\hat{H}}_{\text{JP}} = -(\hat{n}_A + \hat{n}_B)|e\rangle\langle e|_q$, so a wait $\tilde{\tau} = \pi$ is

$$\hat{U}_\pi = |g\rangle\langle g|_q + |e\rangle\langle e|_q\, \hat{\Pi}_{AB}: \quad \hat{U}_\pi|g, n, m\rangle = |g, n, m\rangle, \quad \hat{U}_\pi|e, n, m\rangle = (-1)^{n+m}|e, n, m\rangle. \tag{11}$$

Qubit rotations are $R_y(\pi/2)|g\rangle = (|g\rangle + |e\rangle)/\sqrt{2}$ and $R_y(-\pi/2)(|g\rangle \pm |e\rangle)/\sqrt{2} = |g\rangle$ or $|e\rangle$ (even/odd superposition). Virtual $S^\dagger|g\rangle = |g\rangle$, $S^\dagger|e\rangle = -i|e\rangle$. Write $\lambda = (-1)^{n+m}$ and start from $|g, n, m\rangle$.

Even ($n+m$ even, $\lambda=+1$):

$$\begin{aligned} R_y(+\pi/2)&: \ (|g\rangle+|e\rangle)/\sqrt{2}\,|n,m\rangle \\ \hat{U}_\pi&: \ (|g\rangle+|e\rangle)/\sqrt{2}\,|n,m\rangle \quad \text{(no relative phase)} \\ R_y(-\pi/2)&: \ |g,n,m\rangle \\ S^\dagger&: \ |g,n,m\rangle \\ R_y(+\pi/2),\ \hat{U}_\pi,\ R_y(-\pi/2)&: \ |g,n,m\rangle. \end{aligned}$$

Net: $+|n,m\rangle$, q back in $|g\rangle$.

Odd ($n+m$ odd, $\lambda=-1$):

$$\begin{aligned} R_y(+\pi/2)&: \ (|g\rangle+|e\rangle)/\sqrt{2}\,|n,m\rangle \\ \hat{U}_\pi&: \ (|g\rangle-|e\rangle)/\sqrt{2}\,|n,m\rangle \\ R_y(-\pi/2)&: \ -|e,n,m\rangle \quad \text{(parity now stored on } q\text{)} \\ S^\dagger&: \ i|e,n,m\rangle \quad (-i \text{ on } |e\rangle \text{ times the minus sign}) \\ R_y(+\pi/2)&: \ i(-|g\rangle+|e\rangle)/\sqrt{2}\,|n,m\rangle \\ \hat{U}_\pi&: \ i(-|g\rangle-|e\rangle)/\sqrt{2}\,|n,m\rangle \\ R_y(-\pi/2)&: \ -i|g,n,m\rangle. \end{aligned}$$

Net: $-i|n,m\rangle$, q back in $|g\rangle$.

Together this is $|g\rangle\otimes\hat{U}_{\rm JP}|n,m\rangle$. Local d,e are spectators (echo them if χ_{dA},χ_{eB} must not act).

Duration. Idle time is two waits of π/χ_{qA}. With $\chi_{qA}/2\pi = 1$ MHz,

$$\tau_{\rm JP} = \frac{2\pi}{\chi_{qA}} = 1\ \mu\text{s} \qquad \big(\tilde{\tau}_{\rm JP} = 2\pi \text{ in code}\big). \tag{12}$$

Four transmon $\pi/2$ pulses add ~ 100 ns ($t_q = 24$ ns [2]) and can be treated as instantaneous relative to the waits.

Lindblad equation during the waits (unitless, $\chi_{qA}=\chi_{qB}=1$). Use the *full* 3-qubit, 2-cavity Hamiltonian Eq. (2) so every Fock- and transmon-dependent phase is present:

$$\begin{aligned} \hat{\tilde{H}} = \frac{\hat{H}_{\rm disp}}{\hbar\chi_{qA}} &= \tilde{\omega}_A\hat{n}_A + \tilde{\omega}_B\hat{n}_B + \sum_{j\in\{d,q,e\}} \tilde{\omega}_j|e\rangle\langle e|_j \\ &\quad - \tilde{\chi}_{dA}\,\hat{n}_A|e\rangle\langle e|_d - \tilde{\chi}_{eB}\,\hat{n}_B|e\rangle\langle e|_e - (\hat{n}_A+\hat{n}_B)|e\rangle\langle e|_q \\ &\quad - \tfrac{\tilde{K}_A}{2}\hat{n}_A(\hat{n}_A-1) - \tfrac{\tilde{K}_B}{2}\hat{n}_B(\hat{n}_B-1) - \tilde{K}_{AB}\hat{n}_A\hat{n}_B, \end{aligned} \tag{13}$$

with $\tilde{\chi}_{dA} = \tilde{\chi}_{eB} = 0.033$ and $\tilde{K} \sim 10^{-3}$–10^{-2} ($K/2\pi \sim 1$–10 kHz [4]). The $\tilde{\omega}$ rows are the free phases ($\tilde{\omega}_A = \widetilde{\omega}_A/\chi_{qA} \simeq 5260$ if the cavity is at 5.26 GHz). They can be moved into an interaction picture $\hat{\tilde{H}}_I = e^{i\hat{\tilde{H}}_0\tilde{t}}(\hat{\tilde{H}} - \hat{\tilde{H}}_0)e^{-i\hat{\tilde{H}}_0\tilde{t}}$ with $\hat{\tilde{H}}_0 = \tilde{\omega}_A\hat{n}_A + \tilde{\omega}_B\hat{n}_B + \sum_j \tilde{\omega}_j|e\rangle\langle e|_j$, which removes the stiff GHz numbers without dropping Kerr or any $\chi_{j\mu}$. Every term in Eq. (13) commutes with $\hat{\tilde{H}}_0$ (all are diagonal in the joint Fock–transmon basis), so the interaction picture is exact and time-independent:

$\hat{\tilde{H}}_I = \hat{\tilde{H}} - \hat{\tilde{H}}_0$, i.e. simply delete the $\tilde{\omega}$ rows. The dissipators $\mathcal{D}[\hat{a}]$, $\mathcal{D}[\hat{b}]$, $\mathcal{D}[\hat{\sigma}_-]$, $\mathcal{D}[\hat{\sigma}_z]$ are invariant under this frame change. The system Hamiltonian to put in the Lindblad generator is therefore

$$\boxed{\begin{aligned}\hat{\tilde{H}}_I = &-\tilde{\chi}_{dA}\,\hat{n}_A|e\rangle\langle e|_d - \tilde{\chi}_{eB}\,\hat{n}_B|e\rangle\langle e|_e - (\hat{n}_A+\hat{n}_B)|e\rangle\langle e|_q \\ &- \tfrac{\tilde{K}_A}{2}\hat{n}_A(\hat{n}_A-1) - \tfrac{\tilde{K}_B}{2}\hat{n}_B(\hat{n}_B-1) - \tilde{K}_{AB}\hat{n}_A\hat{n}_B.\end{aligned}} \tag{14}$$

Then

$$\frac{d\hat{\rho}}{d\tilde{t}} = -i\big[\hat{\tilde{H}}_I,\hat{\rho}\big] + \tilde{\kappa}\,\mathcal{D}[\hat{a}]\hat{\rho} + \tilde{\kappa}\,\mathcal{D}[\hat{b}]\hat{\rho} + \sum_{j\in\{d,q,e\}}\Big(\tilde{\gamma}_j\,\mathcal{D}[\hat{\sigma}_-^{(j)}]\hat{\rho} + \tfrac{\tilde{\gamma}_{\phi,j}}{2}\,\mathcal{D}[\hat{\sigma}_z^{(j)}]\hat{\rho}\Big), \tag{15}$$

with $\mathcal{D}[\hat{L}]\hat{\rho} = \hat{L}\hat{\rho}\hat{L}^\dagger - \frac{1}{2}\{\hat{L}^\dagger\hat{L},\hat{\rho}\}$. The $(\hat{n}_A+\hat{n}_B)|e\rangle\langle e|_q$ term implements JP; the local χ_{dA},χ_{eB} and Kerr terms add extra phases on $|n,m\rangle$ and on d,e. Integrate each wait from $\tilde{t}=0$ to π ($\tilde{\kappa}=3.65\times10^{-4}$, $\tilde{\gamma}_d=\tilde{\gamma}_q=\tilde{\gamma}_e=3.18\times10^{-3}$). Cavity loss over the full JP is $\kappa\tau_{\rm JP}\simeq 2.3\times10^{-3}$; transmon T_1 while q spends half the gate in $|e\rangle$ is the larger error.

References

[1] R. Dutta *et al.*, "Solving constrained optimization problems using hybrid qubit–qumode quantum devices," arXiv:2501.11735. Defines the dimensionless noise parameter $\kappa\tau$; per ECD block $\kappa\tau\sim1.5\times10^{-3}$ on the Eickbusch device.

[2] A. Eickbusch *et al.*, *Nat. Phys.* **18**, 1464 (2022). Cavity 5.26 GHz, $T_{1,\rm cav}=436$ μs; transmon 6.65 GHz, $\alpha/2\pi=-181$ MHz, $T_{1,q}\approx50$ μs, $T_2\approx30$ μs; local $\chi_{dA}/2\pi=33$ kHz; $t_q=24$ ns.

[3] Y. Y. Gao *et al.*, *Phys. Rev. X* **8**, 021073 (2018). Two cavities detuned by ~1 GHz; local ancilla plus shared coupler.

[4] C. Wang *et al.*, *Science* **352**, 1087 (2016). Unmatched shared shifts $\chi_A^{ge}/2\pi=0.71$ MHz, $\chi_B^{ge}/2\pi=1.41$ MHz (we instead *match* them).